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SUBJECT CODE NO: - RNP-8007
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group A & B) (NEP) Sem - II
Examination November / December 2025 (To be held in February-2026)
Applied Mathematics-II

[Time: 3 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed.
2) Q.no.1 is compulsory.
3) Solve any four questions from Q.no's 2, 3, 4, 5, 6and 7

SECTION A

Q1 Solve the following questions **20**

1. Bowley's skewness uses which central value?
   a) Mean
   b) Mode
   c) Median
   d) SD

2. If mean = 50 and SD = 0, data is:
   a) Spread
   b) Constant
   c) Skewed
   d) Insignificant

3. If mode is not available, Pearson's skewness =
   a) $\frac{3(\text{Mean Median})}{\text{SD}}$
   b) $\frac{3(\text{Mean Median})}{Variance}$
   c) $\frac{3(\text{Mode Median})}{\text{SD}}$
   d) None

4. For the data: 3, 7, 7, 8, 9, mode is:
   a) 3
   b) 7
   c) 8
   d) 9

5. The equation of tangents to the curve at origin represented by the equation y^2 (2a-x)=x^3 is
   a) x = 0, y=0
   b) x= a, y = 2a
   c) x = 0, y=0
   d) x=2a, y=a

6. The expression for arc length in parametric form is

   a) $ds = \int_a^b xy\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,ds$ b) $ds = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$
   c) $ds = \int_a^b x\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dx$ d) $ds = \int_a^b y\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$

7. The value of $\int_0^\infty e^{-x}x^{n-1}\,dx\; n > 0$ is called as
   a) Beta function
   b) Gamma function
   c) Exponential function
   d) Logarithmic function

8. The value of $\int_0^{\frac{\pi}{2}} sin^4\theta cos^6\,\theta\; d\theta$ is equal to
   a) $\frac{3}{512}$
   b) $\frac{512}{3}$
   c) $\frac{3\,\pi}{512}$
   d) $\frac{512\pi}{3}$

9. The double integral $\iint f(x,y)$ dA represents:

a) Line integral

b) Area under curve

c) Volume under surface

d) None

10. The value of $\int_0^1 \int_0^y dx\ dy$ is

a) 1

b) $\frac{1}{8}$

c) 0

d) $\frac{1}{2}$

11. Area of cardioid r=1-cos θ is:

a) 2π

b) 3π

c) π

d) 4π

12. Volume bounded by x = 0, y = 0, z=x+y and plane z = 0 is:

a) Zero

b) Non-zero

c) Infinity

d) Undefined

SECTION B

Q2 a) Calculate standard Deviation of the following frequency distribution **5**

Weekly expenditure below	5	10	15	20	25
No of students	6	16	28	38	46

5

b) Find Karl-Pearson's coefficient of skewness for the given distribution:

X:	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
F:	2	5	7	13	21	16	8	3

Q3 a) Evaluate $\iint ff\ xy(x + y)\, dxdy$ where A is the area between $=x^2 \& y = x$ **5**

b) Change to polar co-ordinate & evaluate $\int_0^{2a} \int_0^{\sqrt{2x-x^2}} \frac{xdxdy}{x^2+y^2}$ **5**

Q4 a) Evaluate: $x\sqrt{2ax - x2}$ dx **5**

b) Prove that $\beta(m, n) = \int_0^{\infty} \frac{t^{m-1}}{(1-t)^{m+n}} dt$ **5**

Q5 a) Trace the curve $r = a(1 + sin\ \theta)$with full justification **5**

b) Trace the curve $r = asin2\theta$ with justification **5**

Q6 a) Evaluate: $\int_{-1}^{1} \int_0^z \int_{x-z}^{x+z} (x + y + z)\, dxdydz$ **5**

b) Evaluate $\iiint z^2$z2 dxdydz over the volume bounded by the surfaces $x^2 + y^2 = a^2 \&$ $x^2 + y^2 = z, z = 0$ **5**

Q7 a) Find the Q.D.and Its co-efficient **5**

C.I.	20-25	25-30	30-35	35-40	40-45	45-50
f	3	8	16	22	15	5

b) Calculate mean deviation about mean **5**

C.I.	0-10	10-20	20-30	30-40	40-50	50-60	60-70
f	6	5	8	15	7	6	3